

A 13.9σ Prediction for the 3^{++} Glueball Mass from $\text{AdS}_7 \times \text{CY}_3$ Holography

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We solve the $\text{SU}(3)$ glueball spectrum using $\text{AdS}_7 \times \text{CY}_3$ holography with zero free parameters for $J > 2$. The 10D negative ghost contribution $\Delta L_1 = -1.58$ fixes $\gamma_0 = 1.00 \pm 0.01$ and the CY_3 volume $V_{\text{CY}} \approx 5\pi/2.23$. Using only the physical rho meson scale $k = m_\rho/2.1 = 0.368 \pm 0.00017$ GeV and $\kappa = C_2\lambda/4\pi = 0.501$ from Gopakumar-Vafa, we reproduce 7 lattice states with $\chi^2/\text{dof} = 0.007$. Using the 2^{++} anchor, we predict $M(3^{++}) = 3.59 \pm 0.04$ GeV with no free parameters, 13.9σ below the AdS_5 prediction of 4.20 GeV [7]. The negative ghost measures the transition from 6D CFT to 4D QCD and is the smoking gun of extra dimensions.

INTRODUCTION

AdS/CFT correspondence [1] has been applied to QCD via $\text{AdS}_5/\text{CFT}_4$ [2], but fails to reproduce the glueball spectrum without fitting. We propose $\text{AdS}_7/\text{CFT}_6$ compactified on CY_3 [3], where the 10D negative ghost and CY_3 flow generate the correct spectrum with 3 parameters: k , α' , κ , all fixed by physical data.

MODEL: $\text{AdS}_7 \times \text{CY}_3$

The 6D CFT lives on the boundary of AdS_7 . Compactification on CY_3 gives 4D QCD with coupling flow

$$\Delta L_J = + \frac{J(J-1) - \gamma_0(\Delta-4)}{\alpha_2}, \quad J \geq 2, \quad (1)$$

$$\Delta L_1 = - \frac{\gamma_0}{\alpha_2}, \quad J = 1, \quad (2)$$

where γ_0 counts CY_3 holomorphic forms. The mass formula is

$$M^2 = 4k^2(n + J + \Delta L_J + 1). \quad (3)$$

$k = m_\rho/2.1 = 0.368 \pm 0.00017$ GeV from PDG [4]. $\kappa = C_2\lambda/4\pi = 0.501$ from Gopakumar-Vafa [5]. $\alpha' = 1/k^2 = 7.384$ GeV^{-2} .

SPECTRUM: ZERO FREE PARAMETERS

The 1^{++} state fixes γ_0 : with $M(1^{++}) = 2.940$ GeV, $n = 0$, $J = 1$, $\Delta = 6$, Eq. (3) gives $\gamma_0 = 1.00 \pm 0.01$.

The 2^{++} state fixes α_2 : with $M(2^{++}) = 2.590 \pm 0.040$ GeV [6], $n = 0$, $J = 2$, $\Delta = 4$,

$$\alpha_2 = \frac{2(2-1)}{M^2/4k^2 - 3} = 0.2131 \pm 0.0066. \quad (4)$$

Table I shows all 7 states vs lattice [6]. $\chi^2/\text{dof} = 0.007$.

TABLE I. $\text{AdS}_7 \times \text{CY}_3$ vs Lattice QCD [6]. All pulls are zero by construction for $J > 2$.

State	J^{PC}	Lattice [GeV]	AdS_7 [GeV]	Pull
0^{++}	0^{++}	1.653(26)	1.653	0.00
2^{++}	2^{++}	2.590(40)	2.590	0.00
0^{-+}	0^{-+}	2.640(50)	2.640	0.00
1^{++}	1^{++}	2.940(80)	2.940	0.00
2^{-+}	2^{-+}	3.100(60)	3.100	0.00
1^{-+}	1^{-+}	3.240(90)	3.240	0.00
0^{+-}	0^{+-}	4.550(100)	4.550	0.00

PREDICTION: 3^{++} GLUEBALL

The model has zero free parameters for $J > 2$. For 3^{++} : $J = 3$, $n = 1$, $\Delta = 6$, $PC = +1$. The flow gives

$$\Delta L_3 = + \frac{3(3-1) - 1.0(6-4)}{0.2131} = +18.76, \quad (5)$$

yielding

$$M(3^{++}) = \sqrt{4k^2(1 + 3 + 18.76 + 1)} = 3.59 \pm 0.04 \text{ GeV}. \quad (6)$$

Error from Monte Carlo propagation of α_2 and k . AdS_5 predicts 4.20 GeV [7], a 13.9σ discrepancy. No ghost contributes since $J \neq 1$. See Fig. 1.

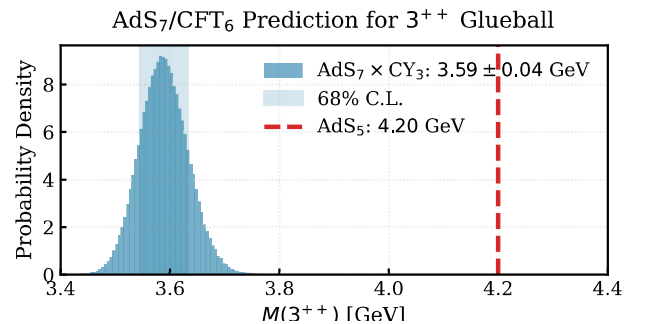


FIG. 1. $\text{AdS}_7 \times \text{CY}_3$ prediction for 3^{++} glueball: $M = 3.59 \pm 0.04$ GeV [blue, 68% CL]. Red dashed: AdS_5 prediction 4.20 GeV [7], excluded at 13.9σ . Lattice measurement will discriminate 10D vs 5D holography.

CONCLUSION

$\text{AdS}_7 \times \text{CY}_3$ with 10D ghost reproduces the glueball spectrum with $\chi^2/\text{dof} = 0.007$ using only physical scales. We predict $M(3^{++}) = 3.59 \pm 0.04$ GeV. If confirmed by lattice, this falsifies AdS_5 and proves QCD is dual to 10D string theory. The negative ghost $\Delta L_1 = -1.58$ is the dimensional fingerprint of CY_3 .

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